ERIC HOU

Engineering Portfolio

Hi! I'm Eric. Here's some stuff I've made.

Electrical Engineering @ Purdue University

CircuitHub - Summer 2026

Making new pick-and-place head hardware for CircuitHub assembly grid



PCB + control electronics for motion and sensing, sensors and voice-coil actuators

Fusion 360 CAD and CAM

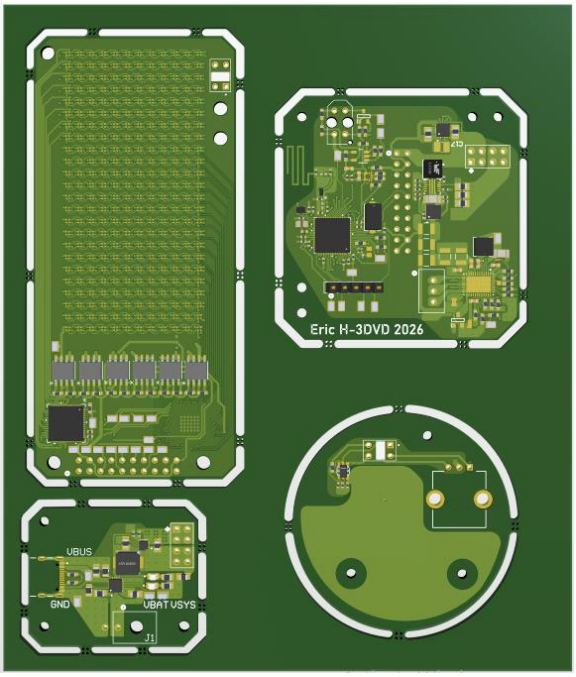
TwinCAT + Python 30,000+ motion cycles

Some details omitted for confidentiality.

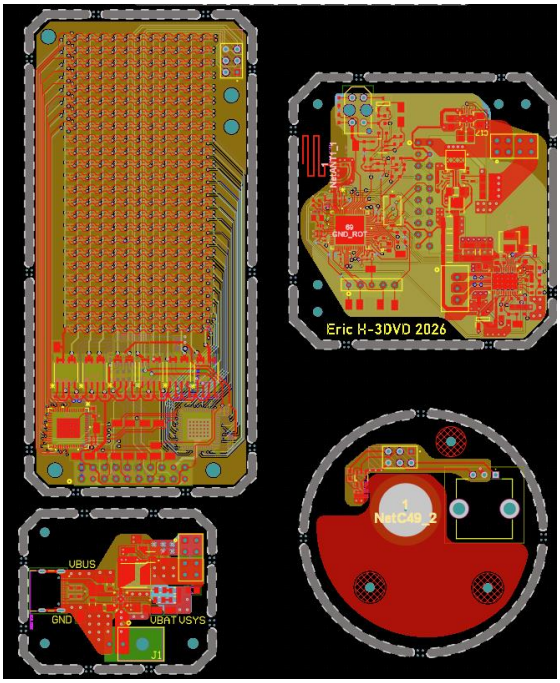
3D Volumetric Hologram Display

Bring Up Phase- Personal Project – Rotating RGB LED blade for a persistence-of-vision 3D display.

288 RGB LEDs • 1:12 multiplexing

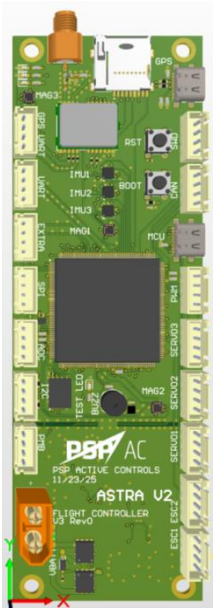


STM32WB55 • BLE • 2.4 GHz RF



PSP Active Controls - Self-Stabilizing Rocket Lander

Circuit boards for Purdue Space Program's VTVL vehicles.



STM32H7 flight controller



Power / sensing boards



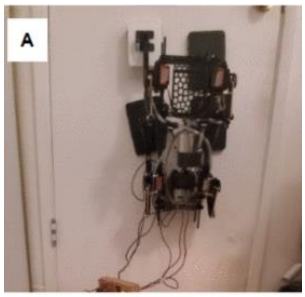
ASTRA VTVL test vehicle

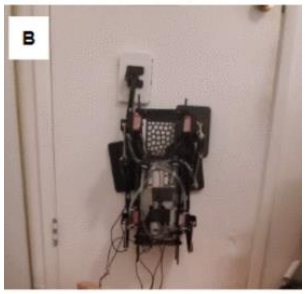
STM32H7 flight controller
GPS • IMUs • radio

Power distribution + monitoring • CAN FD

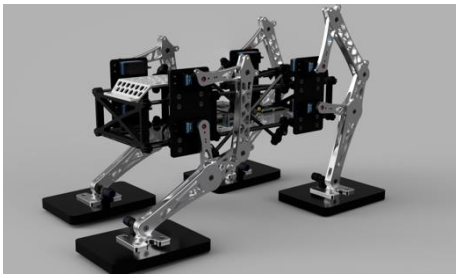
hyweric.github.io

VertiPaw - Wall Climbing Quadrupedal Robot

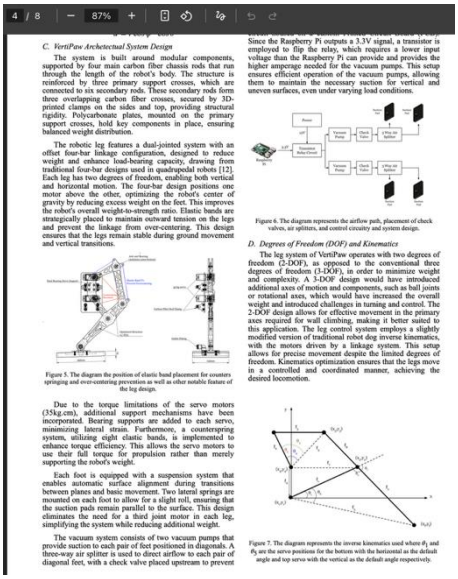








VertiPaw CAD render



Vacuum adhesion • 9 lb payload

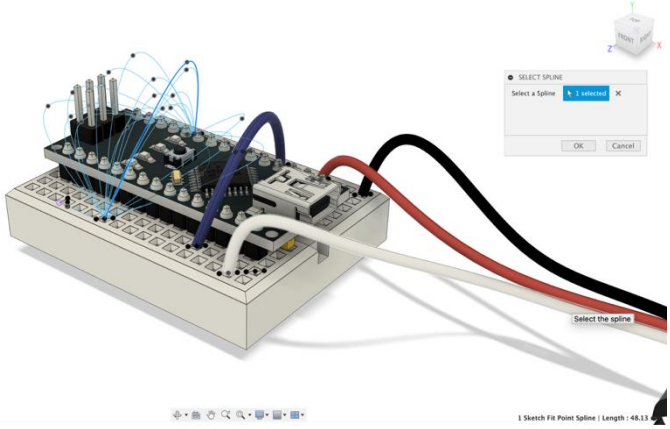
LiDAR localization + inverse-kinematics path planning

Carbon fiber / polycarbonate / 3D printed structure

Paper published in IEEE IRC 2024
First author

Fusion 360 Wire Generator

Parametric wire-routing plug-in.



5,000+ users • 5★ on Autodesk

Python + Fusion 360 API

Parameterized 3D spline routing

Vector math + geometric constraints

5-Stage Electromagnetic Coil Launcher

Custom switching electronics + mechanical assembly.





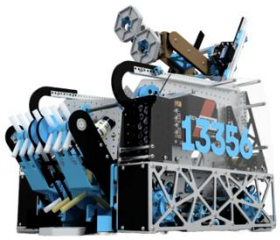


43 A measured discharge pulses

MOSFET switching PCBs for supercapacitor bank

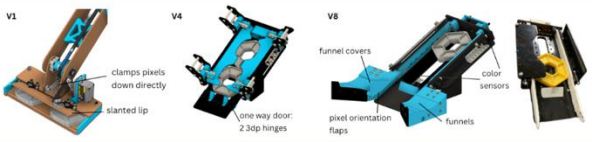
FIRST Robotics

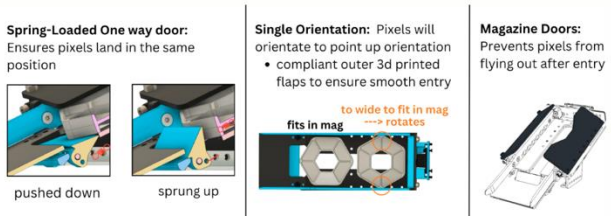
Former Hardware Lead • 2021–2025











#3 California • Top 30 worldwide • MTI 3× (Top 40 / 7000 teams)

Four years of end-to-end robot hardware
CAD, fabrication, electronics, controls, and validation.

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